discussed—in which a fast flyby would be sent to Pluto, the last unexplored planet; but to get there, a gravity assist from the Sun would be employed, in the course of which small probes would actually enter the Sun's atmosphere. And we seem to be on the threshold of a World Consortium for the scientific exploration of Mars. It very much looks as though such projects will be done cooperatively or not at all.

WHETHER THERE ARE VALID, cost-effective, broadly supportable reasons for people to venture to Mars is an open question. Certainly there is no consensus. The matter is treated in the next chapter.

I would argue that if we are not eventually going to send people to worlds as far away as Mars, we have lost the chief reason for a space station—a permanently (or intermittently) occupied human outpost in Earth orbit. A space station is far from an optimum platform for doing science—either looking down at the Earth, or looking out into space, or for utilizing microgravity (the very presence of astronauts messes things up). For military reconnaissance it is much inferior to robotic spacecraft. There are no compelling economic or manufacturing applications. It is expensive compared to robotic spacecraft. And of course it runs some risk of losing human lives. Every shuttle launch to help build or supply a space station has an estimated 1 or 2 percent chance of catastrophic failure. Previous civilian and military space activities have littered low Earth orbit with fast-moving debris—that sooner or later will collide with a space station (although, so far, *Mir* has had no failures from this hazard). A space station is also unnecessary for human exploration of the Moon. *Apollo* got there very well with no space station at all. With *Saturn V* or *Energiya* class launchers, it also may be possible to get to near-earth asteroids or even Mars without having to assemble the interplanetary vehicle on an orbiting space station.

A space station could serve inspirational and educational purposes, and it certainly can help to solidify relations among the spacefaring nations—particularly the United States and Russia. But